



Note

On group-valued potential differences and flows in a signed graph

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ABSTRACT

A signed graph $\Sigma = (G, \sigma)$ is a graph associated with a signature σ on each edge (positive or negative). Let Γ be an additive abelian group and τ be an orientation of Σ . Let $f : E \rightarrow \Gamma$ be a mapping. Then f is a Γ -flow if for each vertex

$$\sum_{(v,e) \in E(v)} \tau(v,e)f(e) = 0,$$

f is a Γ -tension if for every signed circuit $C = v_1 e_1 v_2 e_2 \cdots v_k e_k v_1$,

$$\sum_{i=1}^k \left(-\tau(v_i, e_i) \prod_{j=1}^{i-1} \sigma(e_j) \right) f(e_i) = 0,$$

and f is a Γ -potential difference if f is a Γ -tension such that for every closed walk $W = (v_1 e_1 v_2 e_2 \cdots v_k e_k v_1)$ around an unbalanced circuit,

$$\sum_{i=1}^k f(e_i) \in \{2x : x \in \Gamma\}.$$

We say a mapping $f : E \cup V \rightarrow \Gamma$ is a mixed Γ -flow if for each vertex v ,

$$\sum_{(v,e) \in E(v)} \tau(v,e)f(e) = f(v).$$

Denote by $\mathcal{F}_\Sigma$, $\mathcal{P}_\Sigma$ and $\mathcal{B}_\Sigma$ the sets of Γ -flows, Γ -potential differences, and mixed Γ -flows of Σ respectively.

In this paper, we use a new method to determine the values of $|\mathcal{F}_\Sigma|$ and $|\mathcal{P}_\Sigma|$. Furthermore, we show that $\mathcal{B}_\Sigma/\mathcal{F}_\Sigma$ is isomorphic to $\mathcal{P}_\Sigma$ by a new decomposition of Γ -potential difference.

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1. Introduction

A signed graph is a graph with a signature that assigns a positive or negative sign to each edge. The concept of a signed graph was introduced by Harary [11] to study social networks.

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Tutte [15] initially found that a $\mathbb{Z}_k$ -flow in a plane graph G corresponding to a $\mathbb{Z}_k$ -tension in its dual graph. By this observation, the problem of finding a proper coloring could be converted to the problem of finding a nowhere-zero flow. In 1983, Bouchet [2] found that the orientation for signed graphs is useful to the study of graphs embedded in a non-orientable surface, where nowhere-zero $\mathbb{Z}_k$ -flows emerge as the dual notion to local tensions. Thereafter, the study of nowhere-zero flows in a signed graph has attracted much attention. Recently, the problem of counting the number of nowhere-zero flows in signed graphs was settled by many researchers in [1,8,10,13]. It is different from (ordinary) graphs in that it strongly depends on the order of the group. This means that for the group of odd order, the number of nowhere-zero flows is a polynomial, and for the group of even order this number is another one, the pair of polynomials is called the flow polynomial of a signed graph. Similar to an unsigned graph, the concept of tension was defined in [4] by the orthogonality with signed circuits in a signed graph. Compared to the situation in an ordinary graph, not every tension comes from vertex coloring, only a kind of particular tension called potential-differences is, as defined by Goodall et al. [10]. The method we used in the present paper is simple and essential to demonstrate this variance. More specifically, the variance stems from the fact that the incidence matrix of an oriented unbalanced signed graph is not totally unimodular.

In this paper, some basic preliminaries are given in the second section. In the third section, we use the method of computing the Smith normal form of the incidence matrix of an oriented signed graph to determine the values of $|\mathcal{B}_\Sigma|$ and $|\mathcal{P}_\Sigma|$, which have been obtained in [8,10,13] by different methods. We point out that this method is derived from [3], as remarked in the footnote [1]. As said in [8], the remark is unclear, thus, we provide proof to clarify it in detail.

It is well known that balance is an important concept in the study of signed graphs. Harary's initial description of the balanced signed graph used a unique bipartition. After that, Joglekar et al. [12] gave a characterization of balanced group-labeled graphs that generalized it by assigning abelian group-values to the edge set and vertex set. Moreover, they counted the number of distinct balanced labelings of a graph. The case in directed graphs had been considered by Cherniavsky et al. [6,7]. In the fourth section, inspired by the aforementioned studies we define a mixed flow. As usual, we determine the number $|\mathcal{B}_\Sigma|$ of mixed flows in Σ . Combining the values of $|\mathcal{F}_\Sigma|$ and $|\mathcal{P}_\Sigma|$, we observe that $|\mathcal{B}_\Sigma/\mathcal{F}_\Sigma| = |\mathcal{P}_\Sigma|$. Based on this observation, we find a fundamental directed bonds system to show that $\mathcal{B}_\Sigma/\mathcal{F}_\Sigma$ is isomorphic to $\mathcal{P}_\Sigma$. The fundamental directed bonds are similar to the characteristic vectors determined by frame matroid, but they are different.

2. Preliminaries

The graphs considered in this paper allow parallel edges and loops. As usual, for a graph G , we use $V(G)$ (sometimes write V for short) and $E(G)$ (write E for short) to denote its vertex set and edge set respectively. A *signed graph* $\Sigma = (G, \sigma)$ consists of a graph G together with a *sign function* $\sigma : E(G) \rightarrow \{1, -1\}$. Sometimes, G is also called the underlying graph of Σ .

A *circuit* is a connected 2-regular subgraph in Σ . An *unbalanced circuit* in a signed graph is a circuit that has an odd number of negative edges. A *balanced circuit* is a circuit that is not unbalanced. A signed graph is *unbalanced* if it contains an unbalanced circuit; otherwise, it is *balanced*. A *barbell* is the union of two unbalanced circuits C_1, C_2 and a (possibly trivial) path P with end vertices $v_1 \in V(C_1)$ and $v_2 \in V(C_2)$, such that $C_1 - v_1$ is disjoint from $P \cup C_2$ and $C_2 - v_2$ is disjoint from $P \cup C_1$. We call P the *barbell path* of the barbell. A *signed circuit* is either a balanced circuit or a barbell.

Given a signed graph Σ , *switching* at a vertex v is the inversion of the sign of each edge incident with v . Two signed graphs are said to be *switching-equivalent* if one can be obtained from the other by a series of switchings.

To introduce orienting a signed graph, we begin with the concept of half-edge (see [2,9,17]). Each edge $e = uv$ consists of two half-edges: (u, e) and (v, e) . We associate an oriented half-edge (v, e) with a sign $\tau(v, e)$, where e points toward v if $\tau(v, e) = -1$ and points away v if $\tau(v, e) = 1$. An edge $e = uv$ is oriented in a signed graph based on a sign function τ , ensuring that both (u, e) and (v, e) are oriented subject to $\tau(u, e)\tau(v, e) = -\sigma(e)$. Each edge in Σ has two possible orientated orientation. For a vertex v in an orientated signed graph, we denote by $E(v)$ the set of all half-edges (v, e) .

Let (Σ, τ) be an oriented signed graph. Let Γ be an additive Abelian group. A map $\mathbf{f} : E \rightarrow \Gamma$ is called a Γ -flow if the usual conservation law (Kirchhoff's law) is satisfied, that is, for each vertex v ,

$$\sum_{(v,e) \in E(v)} \tau(v, e)\mathbf{f}(e) = 0.$$

A walk on G is a sequence $W = v_0e_1v_1e_2v_2 \dots e_{n-1}v_{n-1}e_nv_n$ of vertices and edges such that the vertices v_i, v_{i-1} are endpoints of the edge $e_i, i = 1, \dots, n$. The vertex v_0 and the vertex v_n are called the initial vertex and the terminal vertex of W , respectively. If $v_0 = v_n$, W is a closed walk. A walk is positive if it has an even number of negative edges; otherwise, it is negative.

A mapping $\mathbf{t} : E \rightarrow \Gamma$ is called a Γ -tension [4,10] if, for any balanced circuit or barbell $C = v_1e_1v_2e_2 \dots v_ke_kv_1$, the following equation holds:

$$\sum_{i=1}^k \left(-\tau(v_i, e_i) \prod_{j=1}^{i-1} \sigma(e_j) \right) \mathbf{t}(e_i) = 0. \quad (1)$$

where the summation runs over edges of a positive closed walk with respect to a balanced circuit or barbell $C = v_1e_1v_2e_2 \dots v_ke_kv_1$, so if e is on the barbell path it contributes to the sum two times. Moreover, a Γ -tension $\mathbf{t}$ is said

to be a Γ -potential difference of Σ [10] if for every closed walk $W = (v_1 e_1 v_2 e_2 \cdots v_k e_k v_1)$ around an unbalanced circuit,

$$\sum_{i=1}^k \mathbf{t}(e_i) \in 2\Gamma \quad (2)$$

where $2\Gamma = \{2x : x \in \Gamma\}$.

The incidence matrix of the oriented signed graph (Σ, τ) is a matrix $\mathbf{M}(\Sigma, \tau) = [\mathbf{m}(v, e)]$, where $(v, e) \in (V, E)$, and the (v, e) -entry is defined by

$$\mathbf{m}(v, e) = \begin{cases} \tau(v, e), & \text{if } e \text{ is a non-loop at } v; \\ 2\tau(v, e), & \text{if } e \text{ is a negative loop at } v; \\ 0, & \text{otherwise} \end{cases}$$

For simplicity, we write $\mathbf{M}$ for $\mathbf{M}(\Sigma, \tau)$. We give an equivalent definition, an Γ -flow on (Σ, τ) is a function $f : E \rightarrow \Gamma$ such that $\mathbf{M}f = 0$, i.e., f is in null space of $\mathbf{M}$.

Let f be a flow with respect to an orientation. If we switch the direction of an edge e of Σ and replace $f(e)$ by its additive inverse, we obtain a new flow f' corresponding to the new orientation. Consequently, the number of flows (nowhere-zero) in two different orientations of Σ are always equal. Since this property also holds to the other functions (tension, potential-difference, and mixed flow) discussed in this paper, we count these numbers for a given orientation.

Let R be a principal ideal domain. Given an $m \times n$ matrix M over R , M can be put into Smith normal form by elementary row and column operations:

- (i) interchanging two rows (columns),
- (ii) multiplying one row (respectively, column) by units of R ,
- (iii) adding a multiple of one row (respectively, column) to another (respectively, column).

The Smith normal form is a matrix with r non-zero diagonal elements $d_1, \dots, d_r$ and all other entries zero, where d_i divides d_{i+1} for $i = 1, \dots, r-1$. The elements $d_1, \dots, d_r$ are uniquely determined up to multiplication by units of R . They are the invariant factors of M . By convention, we also take 0 to be an invariant factor with multiplicity $n-r$, so that there are n invariant factors in all. Recall that a minor of a matrix is a determinant of some square submatrix. As is well known that not every $k \times k$ minor can be preserved by elementary operations, however, the greatest common divisor (gcd) of all $k \times k$ minors should be. Denote D_i to be the gcd of all $i \times i$ minors. Therefore, we can compute the invariant factor by $d_i = D_i/D_{i-1}$ and let $D_0 = 1$, for $i = 1, \dots, r$ [14, Theorem 2.4].

We use the following notation for submatrices of a matrix M : for the set $I \subseteq \{v_1, v_2, \dots, v_n\}$ and the set $J \subseteq \{e_1, e_2, \dots, e_m\}$,

$M[I; J]$ denotes the submatrix of M whose rows are indexed by I and columns are indexed by J .

$M(I; J)$ denotes the submatrix of M obtained by removing the rows indexed by I and removing the columns indexed by J .

3. The number of potential differences and flows

This section is devoted to counting the potential differences and flows in an unbalanced signed graph. For completeness, some easy lemmas are presented in detail.

A unicycle graph, or unicycle for short, is a connected graph with an equal number of edges and vertices. Let $\Sigma - v$ be a signed graph obtained from Σ by removing vertex v and all its incident edges.

Lemma 3.1. *If (Σ, τ) is an oriented unbalanced (resp. balanced) unicycle, then the determinant of its incidence matrix is ± 2 (resp. 0).*

Proof. Let (Σ, τ) be an oriented unbalanced graph with incidence matrix $\mathbf{M}$ and t vertices not on the circuit. We prove the statement by induction on t . If $t = 0$, it is easy to find the elementary matrices P and Q , such that

$$M' = PMQ = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 & a \\ -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{bmatrix}_{n \times n}.$$

and, both matrices P and Q are corresponding to permutation and multiplying -1 . By a cofactor expansion across the last row,

$$\det(\mathbf{M}) = \pm \det(M') = \pm((-1)^{2n} + (-1)^{2n}a).$$

If Σ is unbalanced (resp. balanced), $a = 1$ (resp. $a = -1$), we have $\det(\mathbf{M}) = \pm 2$ (resp. 0). If $t \geq 0$, assume the statement holds for some t . Let (Σ, τ) be an oriented unbalanced unicycle with $t + 1$ vertices not on the circuit, and p is a pendant

vertex (degree of 1). Without loss of generality, assume p is a tail of e . Then we have a cofactor expansion across the row corresponding to vertex p and get

$$\det(\mathbf{M}) = 1 \cdot (\pm \mathbf{M}(p; e)). \quad (3)$$

Because $\pm \mathbf{M}(p; e)$ is the incident matrix of $\Sigma - p$, by induction hypothesis, the statement holds. $\square$

By a similar argument, we have the following lemma.

Lemma 3.2. *Let (H, τ) be an oriented tree with at least one edge and $\mathbf{M}$ be the incidence matrix of H . Then $\det(\mathbf{M}(i;)) = \pm 1$ for all vertices i of H .*

Lemma 3.3 ([16]). *Let (Σ, τ) be an oriented signed graph with n vertices. For some $n \times n$ minor $\mathbf{M}[S]$ in $\mathbf{M}$, $\det(\mathbf{M}[S])$ is either 0 or $\pm 2^l$, where l is a non-negative integer no more than n .*

Proof. Let $\mathbf{M}$ be an incidence matrix of an oriented signed graph (G, τ) . Let S be a subgraph generated by the vertex set I and edge set J , and $|I| = |J| = n$. S may contain a loose edge, which lacks an end-vertex. If S has an isolated vertex or a loose edge, then $\det(\mathbf{M}[I, J]) = 0$.

Therefore, we assume S has no isolated vertices and loose edges, and is composed of t connected components $S_1, S_2, \dots, S_t$. Note that $|V(S_i)| - |E(S_i)| \leq 1$, for each $1 \leq i \leq t$. If S has a unique connected component, then S must be a unicycle. By Lemma 3.1, $\det(\mathbf{M}[S]) = 0$ or $\det(\mathbf{M}[S]) = \pm 2$ determined by the circuit in S being balanced or not. Now we consider $t \geq 2$. Assume $|V(S_i)| - |E(S_i)| = 1$ for some i , which means S_i is a tree. We can recursively remove a pendant vertex and its incident edge from S_i until only an isolated vertex v remains. This process implies that $\det(\mathbf{M}[S]) = \pm \det(\mathbf{M}(V(S_i) \setminus \{v\}; E(S_i)))$. Then $\det(\mathbf{M}[S]) = 0$ because the row corresponding to v in $\mathbf{M}(V(S_i) \setminus \{v\}; E(S_i))$ is only consist of zero. Since $|V(S_i)| < |E(S_i)|$ gives $\det(\mathbf{M}[S]) = 0$, we only need to deal with the case of $|V(S_i)| = |E(S_i)|$. Then in this case, we observe that S_i is either a tree that removes a pendant vertex and retains its pendant edge or a unicycle. Consequently, by Lemma 3.1, Lemma 3.2 and $\det(\mathbf{M}[S]) = \prod_{i=1}^t \det(\mathbf{M}[S_i])$, we have $\det(\mathbf{M}[S]) = \pm 0^{b(S)} 2^{u(S)}$, $b(S)$ is the number of balanced unicycle and $u(S)$ is the number of unbalanced unicycle in S . This completes the proof. $\square$

In the following, when computing the Smith normal form of $\mathbf{M}$, we should specialize it to the integral domain $\mathbb{Z}$.

Theorem 3.4. *Let Σ be a connected and oriented unbalanced signed graph and $\mathbf{M}$ be its incidence matrix, then the Smith normal matrix $S_{\mathbf{M}}$ of $\mathbf{M}$ is*

$$S_{\mathbf{M}} = \begin{bmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & 2 & \\ & & & & \ddots \end{bmatrix}_{n \times m}.$$

Proof. Instead of finding two invertible matrices $P_{n \times n}$ and $Q_{m \times m}$ such that $PMQ = S_{\mathbf{M}}$, we obtain the statement by determining its invariant factors.

At first, we claim that the first $n - 1$ invariant factors are 1 (up to sign). For any tree T with $i \leq n - 1$ edges and v is a vertex of T , using Lemma 3.2 we have $\det(\mathbf{M}_{\mathbf{T}}(v;)) = \pm 1$, where $\mathbf{M}_{\mathbf{T}}$ is the incidence matrix of T . According to the definition of D_i , we have $D_i = \pm 1$ for $i = 1, \dots, n - 1$. At last, we show that $D_n = \pm 2$. Let T be a spanning tree in Σ . Since Σ is unbalanced, we can find an edge e such that $T' = T + e$ contains an unbalanced circuit C . By Lemma 3.1 we have $\det \mathbf{M}_{T'} = \pm 2$. Moreover, using Lemma 3.3 we have $D_n = \pm 2$. Since invariant factors $d_i = D_i/D_{i-1}$ are uniquely determined up to multiplication by -1 (unit of $\mathbb{Z}$), we achieve the desired result. $\square$

The following results can be found in our paper [13] and Goodall et al. [10].

Corollary 3.5. *Let Γ be an additive Abelian group and let Σ be a connected unbalanced signed graph with n vertices and m edges. Then the number of Γ -potential differences of a signed graph Σ is given by*

$$|\mathcal{P}_{\Sigma}| = |\Gamma|^{n-1} |2\Gamma|. \quad (4)$$

The number of Γ -flows is given by

$$|\mathcal{F}_{\Sigma}| = |\Gamma|^{m-n+1} \left(\frac{1}{|2\Gamma|} \right). \quad (5)$$

Proof. Applying elementary row and column operations to $\mathbf{M}$ does not change the number of solutions of these equations $\mathbf{M}f = 0$ and the number of linear combinations of vectors generated by the rows of $\mathbf{M}$: row operations leave the solution

set and row vectors unaltered, while column operations can be viewed as applying an invertible transformation to it. Since a Γ -flow is a solution of $\mathbf{M}\mathbf{f} = 0$ and a Γ -potential difference is a linear combination of rows in $\mathbf{M}$ [10, Theorem 7.14], their numbers are equal to the numbers of ones corresponding to $S_{\mathbf{M}}$. Recalling Theorem 3.4, the smith normal form $S_{\mathbf{M}}$ corresponding the equations are $x_1 = 0, \dots, x_{n-1} = 0, 2x_n = 0$, with $x_{n+1}, \dots, x_m$ arbitrary. So the Eqs. (4) and (5) are obtained. $\square$

Remark 1. The matrix perspective offers several advantages. First, it allows for the easy understanding that the number of potential differences and flows remains unchanged under switching inversion. Second, in an unsigned graph, every tension $\mathbf{t}$ can be derived from a coloring $\mathbf{c}$ by assigning an oriented edge $\vec{uv}$ the value of $\mathbf{c}(u) - \mathbf{c}(v)$. Cameron [3] provides a necessary and sufficient condition for all tensions produced this way: the matrix $\mathbf{M}$ must be totally unimodular. However, as established in Theorem 3.4, this does not hold for unbalanced signed graphs, where only a specific class of tensions can be realized. Goodall et al. [10] identify this class as Γ -potential differences. Additionally, using Theorem 3.4, we can easily derive the torsion of $\mathbf{M}$ as defined by Chen et al. [5]. While computing the Smith normal form is an effective method for counting Γ -potential differences and Γ -flows, it does not clearly illustrate their structure. Thus, we provide a decomposition of Γ -flows using fundamental circuits [13].

4. The characterization potential-difference by quotient flow

We begin with a definition motivated by the thoughts in [6,7,12]. Then the relationship between it and potential differences will be established in this section.

Definition 1. A function $\mathbf{f} : E \cup V \rightarrow \Gamma$ is called a *mixed Γ -flow* if for each vertex v , the following equation holds:

$$\sum_{(v,e) \in E(v)} \tau(v,e)\mathbf{f}(e) = \mathbf{f}(v).$$

Recall that $\mathcal{F}_{\Sigma}$, $\mathcal{P}_{\Sigma}$ and $\mathcal{B}_{\Sigma}$ are the sets of Γ -flows, Γ -potential differences, and mixed Γ -flows of Σ respectively. We first give an easy proposition without proof.

Proposition 4.1. Let Σ be a signed graph with m edges and Γ be an abelian group, then

$$|\mathcal{B}_{\Sigma}| = |\Gamma|^m.$$

Viewing $\mathcal{B}_{\Sigma}$ and $\mathcal{F}_{\Sigma}$ as the set of vectors forms two abelian groups with respect to vector-addition. Since $\mathcal{F}_{\Sigma}$ is a subgroup of $\mathcal{B}_{\Sigma}$, then $\mathcal{B}_{\Sigma}/\mathcal{F}_{\Sigma}$ is a quotient group. For an unbalanced signed graph, by Eqs. (4) and (5) one has an interesting fact that

$$|\mathcal{B}_{\Sigma}/\mathcal{F}_{\Sigma}| = |\mathcal{P}_{\Sigma}|.$$

So a natural question is raised. Can we find a bijection, or an isomorphism, between $\mathcal{B}_{\Sigma}/\mathcal{F}_{\Sigma}$ and $\mathcal{P}_{\Sigma}$? We answer it in the rest of this paper. In the following, we give a decomposition of Γ -potential difference by the indicator vector of fundamental directed bonds.

Let T be a spanning tree of the underlying graph G . Then we can make all edges from T positive by switching operations. We choose an arbitrary negative edge e_0 and call $T_0 = T + e_0$ a *signed rooted tree* of Σ with root edge e_0 . Let $\bar{T}_0 = E(\Sigma) \setminus E(T_0)$. For a signed graph Σ , and $X \subseteq V(\Sigma)$, we denote by $[X, X^C]$ the set of edges between X and its complement X^C , by $\Sigma[X]$ the subgraph of Σ induced by X , and by $E(X)$ the set of the edges in $\Sigma[X]$. A non-empty edge subset $B \subseteq E(\Sigma)$ is called a *cut* Σ if it has the form $B = [X, X^C] \cup E_X$, where $X \subseteq V(\Sigma)$ is non-empty and $E_X \subseteq E(X)$ is minimal to have $\Sigma[X] - E_X$ balanced. A cut is called a *bond* of Σ if it is minimal. For any $e \in T$, let B_e be the bond $[X, X^C] \cup E_X$, where X and X^C are the vertex subsets of the two components in $T - e$ and, without loss of generality we assume that $e_0 \in E_X$. We call B_e a *fundamental bond*. Furthermore, let B_{e_0} be the particular bond $[V(\Sigma), \emptyset] \cup E_{V(\Sigma)}$, in which $E_{V(\Sigma)}$ is consist of all negative edges in $E(\Sigma) \setminus E(T)$. We note that, by the minimality of a bond, if X^C is balanced then $[X, X^C]$ is a bond but $[X, X^C] \cup E_X$ is not. In this case, we let $B_e = [X^C, X]$.

Let τ be a fixed orientation of Σ . For an edge $e = uv$, we also write $\tau(u, e)$ and $\tau(v, e)$ together as $\tau(e)$ for short. For a fundamental bond B_e with $e = uv \neq e_0$ and $u \in X$, $v \in X^C$, the *fundamental directed bond* $\vec{B}_e$ of G is an orientation τ' of B_e such that $\tau'(e) = \tau(e)$, $\tau'(x, xx') = \tau(u, e)$ for every $xx' \in [X, X^C]$ with $x \in X$, $x' \in X^C$, and $\tau'(y, yy') = -\tau(u, e)$ for every negative edge $yy' \in E[X]$. For $e \in T$, let B_e^{τ} be the orientation τ restricted to B_e . We associate with B_e a function $\mathbf{t}_e$ defined by

$$\mathbf{t}_e(a) = \begin{cases} 1, & \text{if } a \in \vec{B}_e \text{ and } a \in [X, X^C]; \\ -1, & \text{if } a \in B_e^{\tau} \setminus \vec{B}_e \text{ and } a \in [X, X^C]; \\ 2, & \text{if } a \in \vec{B}_e \text{ and } a \in E_X; \\ -2, & \text{if } a \in B_e^{\tau} \setminus \vec{B}_e \text{ and } a \in E_X; \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

For e_0 , we associate with B_{e_0} a function $\mathbf{s}$ on $E(\tau)$ defined by

$$\mathbf{s}(e) = \begin{cases} 1, & \text{if } e \in E_{V(\Sigma)} \text{ and } \tau(e) = \tau(e_0); \\ -1, & \text{if } e \in E_{V(\Sigma)} \text{ and } \tau(e) = -\tau(e_0); \\ 0, & \text{otherwise.} \end{cases} \quad (7)$$

For convenience, in addition to being the functions on E we also regard $\mathbf{t}_e$ and $\mathbf{s}$ as the vectors indexed by the edges of (Σ, τ) .

Theorem 4.2. Let Γ be an additive Abelian group and let Σ be a connected unbalanced signed graph. Let T be a spanning tree of Σ consisting of positive edges and let $e_0 \in E_{V(\Sigma)}$. Then the class of all Γ -potential differences in G is given by

$$\mathcal{P}_\Sigma = \{\gamma\mathbf{s} + \sum_{e \in T} \gamma_e \mathbf{t}_e : \gamma \in 2\Gamma, \gamma_e \in \Gamma\}. \quad (8)$$

Proof. It can be seen [4] that $\mathbf{s}$ and $\mathbf{t}_e$ are the cut characteristic vectors of B_{e_0} and B_e , respectively. Following [4, Lemma 3.3], $k\mathbf{s}$ and $k_e \mathbf{t}_e$ are integral tensions for any integers k and k_e . Since Γ is an additive group, $\gamma\mathbf{s}$ and $\gamma_e \mathbf{t}_e$ are tensions for any $\gamma \in 2\Gamma$ and $\gamma_e \in \Gamma$. Thus, $\gamma\mathbf{s} + \sum_{e \in T} \gamma_e \mathbf{t}_e$ is a Γ -tension.

Now we will prove that, if $\gamma \in 2\Gamma$ then

$$\gamma\mathbf{s} + \sum_{e \in T} \gamma_e \mathbf{t}_e \quad (9)$$

is a Γ -potential difference. Let $W = v_1, e_1, v_2, e_2, \dots, v_k, e_k, v_1$ be an arbitrary closed walk around an unbalanced circuit C in Σ .

Firstly, we consider the Γ -tension $\gamma\mathbf{s}$. By (7), every negative edge on W is assigned the value γ or $-\gamma$ by $\gamma\mathbf{s}$, while other edges on W are assigned zero. Therefore, the total values of the edges on W must be in 2Γ as $\gamma \in 2\Gamma$. Then by (2), $\gamma\mathbf{s}$ is a Γ -potential difference.

Secondly, we show that $\gamma_e \mathbf{t}_e$ is a Γ -potential difference for any $e \in T$. If W is contained in $G[X^c]$, then by (6) every edge is assigned zero and, therefore the total value is zero. If W is contained in $G[X]$, then again by (6), every negative edge on W is assigned $2\gamma_e$ or $-2\gamma_e$, while other edges are assigned zeros. The total value is in 2Γ in this case. Finally, if $W \cap [X, X^c] \neq \emptyset$, then the number of edge in $W \cap [X, X^c]$ is even as C is a circuit. Since every edge in $W \cap [X, X^c]$ is assigned γ_e or $-\gamma_e$, the total values of such edges is $2q\gamma_e$ for some integer q . On the other hand, the total value of other edges as has been proved above, is in 2Γ . Hence, by (2), $\gamma_e \mathbf{t}_e$ is a Γ -potential difference. As a result, $\gamma\mathbf{s} + \sum_{e \in T} \gamma_e \mathbf{t}_e$ is a Γ -potential difference.

Conversely, let $\mathbf{t}$ be an arbitrary Γ -potential difference in G . We need to prove that $\mathbf{t}$ can be written as the combination (9) with $\gamma \in 2\Gamma$.

In (9), we set $\gamma_e = \mathbf{t}(e)$ and $\gamma = \mathbf{t}(e_0) - \sum_{e \in T} \gamma_e \mathbf{t}_e(e_0)$. Let C'_0 be the edge set of the unique unbalanced circuit in T_0 and let $C_0 = C'_0 \setminus \{e_0\}$. Then

$$\gamma = \mathbf{t}(e_0) - \sum_{e \in C_0} \gamma_e \mathbf{t}_e(e_0) - \sum_{e \in T - C_0} \gamma_e \mathbf{t}_e(e_0).$$

By (6), we have $\gamma_e \mathbf{t}_e(e_0) \in 2\Gamma$ for any $e \in T - C_0$. On the other hand, since $C'_0 = C_0 \cup \{e_0\}$ is the edge set of the unique unbalanced circuit in T_0 and $\mathbf{t}$ is a potential difference, we have $\sum_{e \in C'_0} \mathbf{t}(e) \in 2\Gamma$, which is equivalent to that

$$\mathbf{t}(e_0) - \sum_{e \in C_0} \gamma_e \mathbf{t}_e(e_0) \in 2\Gamma.$$

Therefore, $\gamma \in 2\Gamma$.

Let

$$\mathbf{t}' = \mathbf{t} - (\gamma\mathbf{s} + \sum_{e \in T} \gamma_e \mathbf{t}_e). \quad (10)$$

For any $e \in T$, by the definition of the vector $\mathbf{s}$ we have $\mathbf{s}(e) = 0$ since e is positive. Notice that $\mathbf{t}_e(e) = 1$ and $\mathbf{t}_e(e') = 0$ for any $e, e' \in T$ with $e \neq e'$. So by (10), we have $\mathbf{t}'(e) = \mathbf{t}(e) - (\gamma\mathbf{s}(e) + \gamma_e \mathbf{t}_e(e)) = 0$ as $\gamma_e = \mathbf{t}(e)$. Furthermore, by (7) we have $\mathbf{s}(e_0) = 1$. Thus, again by (10), we have

$$\mathbf{t}'(e_0) = \mathbf{t}(e_0) - (\gamma\mathbf{s}(e_0) + \sum_{e \in T} \gamma_e \mathbf{t}_e(e_0)) = 0$$

since $\gamma = \mathbf{t}(e_0) - \sum_{e \in T} \gamma_e \mathbf{t}_e(e_0)$. As a result, we have $\mathbf{t}'(e) = 0$ for every edge $e \in T_0$. We claim that $\mathbf{t}'(e) = 0$ for every edge $e \in \bar{T}_0$ (This was also observed in [4, 10]). We consider the unique signed circuit C_e in $T_0 + e$. Since $\mathbf{t}'$ is a tension and $\mathbf{t}'(e') = 0$ for any $e' \in C_e - e \subseteq T_0$, we must have $\mathbf{t}'(e) = 0$ and, therefore $\mathbf{t}' = \mathbf{0}$ (the vector of all zeros). In conclusion, $\mathbf{t}$ is represented as the combination (9) and $\gamma \in 2\Gamma$. Theorem 4.2 follows. $\square$

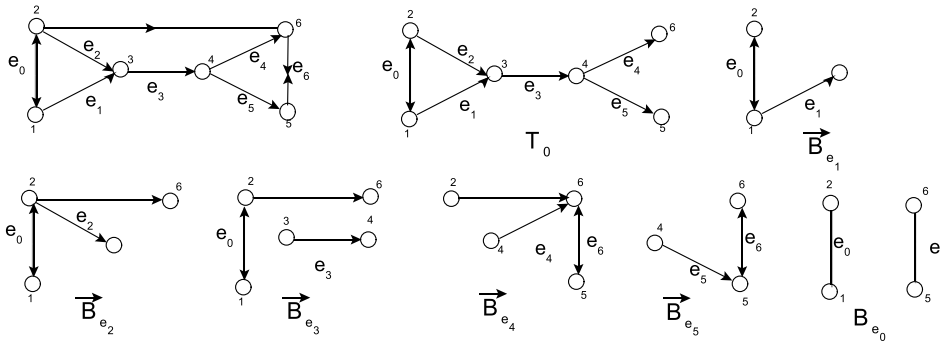


Fig. 1. An example for the fundamental directed bonds system.

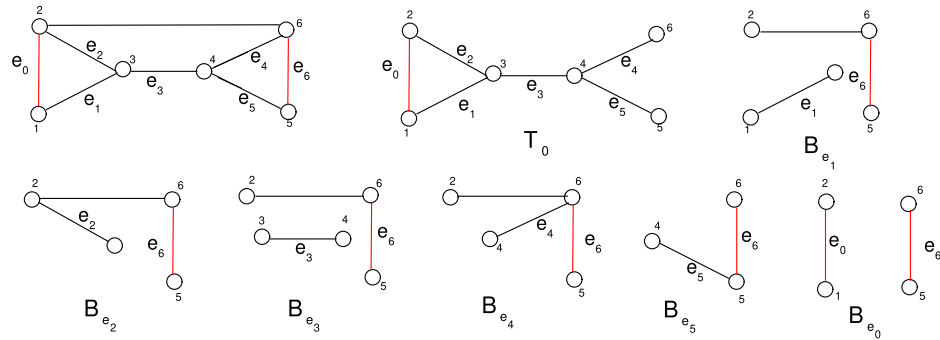


Fig. 2. The fundamental bonds with respect to a certain base T_0 determined by frame matroid.

Theorem 4.3. Let Σ be a connected unbalanced signed graph, there is an isomorphism between $\mathcal{B}_\Sigma/\mathcal{F}_\Sigma$ and $\mathcal{P}_\Sigma$.

Proof. Let Σ be an unbalanced graph with orientation τ . And let $\mathcal{F} = \{(x_1, x_2, \dots, x_n) : x_1 + x_2 + \dots + x_n \in 2\Gamma\}$, then $\mathcal{F}$ is a group with respect to vector addition. Using double-counting, for any $\mathbf{f} \in \mathcal{B}_\Sigma$, we have

$$\sum_{v \in V(G)} \mathbf{f}(v) = \sum_{v \in V(G)} \left(\sum_{(v, e) \in E(v)} \tau(v, e) \mathbf{f}(e) \right) = \sum_{e \in E} \left(\sum_{v \in V(G)} \tau(v, e) \mathbf{f}(e) \right),$$

which implies that

$$\sum_{v \in V(G)} \mathbf{f}(v) \in 2\Gamma.$$

Let $\mathbf{f}_V$ denote $\mathbf{f}$'s restriction on vertex set V , and θ is a map defined as $\theta(\mathbf{f}) = \mathbf{f}_V$. It is easily verified that θ is an isomorphism between $\mathcal{B}_\Sigma/\mathcal{F}_\Sigma$ and $\mathcal{F}$. Now it is sufficient to find an isomorphism between $\mathcal{F}$ and $\mathcal{P}_\Sigma$. Assume $\mathbf{X}_\gamma = (x_1, x_2, \dots, x_n) \in \mathcal{F}$, and $\sum_{i=1}^n x_i = \gamma \in 2\Gamma$. Recall $\mathcal{P}_\Sigma = \{\gamma \mathbf{s} + \sum_{e \in T} \gamma_e \mathbf{t}_e : \gamma \in 2\Gamma, \gamma_e \in \Gamma\}$ and define a map μ as

$$\mathbf{X}_\gamma \rightarrow \gamma \mathbf{s} + \sum_{e_i \in T} x_i \mathbf{t}_{e_i}.$$

It is also an isomorphism, completed. $\square$

Remark 2. We provide a decomposition of the Γ -potential difference to establish an isomorphic map between $\mathcal{B}_\Sigma/\mathcal{F}_\Sigma$ and $\mathcal{P}_\Sigma$. Alternatively, one can utilize the characteristic vectors associated with bonds defined in [4] to achieve a similar isomorphism. To highlight the distinction between our fundamental directed bonds and those defined by frame matroid [4], refer to Fig. 1 (where red lines represent negative edges) and Fig. 2. We believe the decomposition in Theorem 4.2 may have broader applications.

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Data availability

No data was used for the research described in the article.

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